Mini Seach Engine

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Description

Mini Search Engine is a search platform that crawls and indexes documents, processes user queries, and ranks results using TF-IDF scoring. The project combines a Python and Flask backend with a React frontend to provide fast and relevant search results through an interactive web interface.

Backend

- Python
- Flask

Frontend

- React
- HTML/CSS

Search Technologies

- Inverted Index
- TF-IDF ranking

Discoveries & Difficulties

Discoveries

- Learned how search engines organize data using an inverted index
- Realized indexing greatly improves search speed compared to scanning every document
- Learned how web crawlers collect and process webpage data
- Gained experience connecting Flask with a React frontend

Difficulties

- Making searches fast with large amounts of data
- Correctly ranking the most relevant search results
- Handling duplicate or irrelevant results
- Debugging communication between Flask API and React frontend
- Organizing indexed data efficiently
- Processing and cleaning text data from crawled webpages
- Managing errors when webpages failed to load during crawling

Demo: